

The performance impact of post-M&A interdepartmental integration: An empirical analysis

Chi-Hsiang Chen^{a,*}, Yong-Yang Chang^{b,1}, Ming-ji James Lin^c

^a Department of Business and Entrepreneurial Management, Kainan University, Taiwan ROC No.1 Kainan Road, Luzhu Shiang, Taoyuan 33857, Taiwan

^b Department of Business and Entrepreneurial Management, Kainan University, Taiwan ROC

^c Department of Business Administration, National Central University, Taiwan ROC

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ABSTRACT

This study investigates how post-M&A interdepartmental integration affects the efficiency and effectiveness of new product development (NPD). The total sample size was 251 respondents. Confirmatory factor analysis (CFA) and structural equation modeling (SEM) were used for statistical analysis. Analytical results indicate that, although collaboration interdepartmental integration positively correlates with product vision, interaction interdepartmental integration does not reach a significant level. Despite the fact that some interaction is essential when developing new product competitive advantage (NPCA), such interaction does not necessarily achieve success. Further, product vision positively correlates with new product competitive advantage and NPD performance, and new product competitive advantage positively correlates with NPD performance. In addition, an examination of the mediation effect in terms of *Sobel t*-test reveals that product vision is a significant mediator for the influence of interdepartmental integration on new product competitive advantage, while the new product competitive advantage is also a significant mediator for the influence of interdepartmental integration on NPD performance. Moreover, this study provides a framework for managing post-M&A integration and closes with a discussion of the theoretical and practical implications of the research findings.

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1. Introduction

Rapid technological advances, shortened product life cycles, and increased technological complexity are increasingly compelling firms to outsource technological development (Bannert & Tschirky, 2004). A major reason for undertaking a merger and acquisition is to access technological knowledge and enhance new product development (NPD) capabilities (Grimpe, 2007). Immediately after an acquirer completes the legal procedures of an M&A, achieving the so-called M&A synergy of $1 + 1 > 2$ is essential for effective interdepartmental integration. However, existing empirical research shows that most M&As fail to achieve the desired results; in fact, the failure rate is roughly 60%–80% (Mark & Mirvis, 2001). Identifying ways to increase the success rate of M&As is a major objective of academia and the industry (Jaspers & Ende, 2006; Palmatier, Miao, & Fang, 2007).

New products form the core of enterprise sustainability. However, NPD is an intricate and complex process. Each step of this process – from investigating consumer needs, interdepartmental cooperation, technical risks during the development process, analyzing strategies

of competitors, obtaining resources, and others – significantly impacts NPD. New product development (NPD) represents the focal point of competition for many firms (Brown & Eisenhardt, 1995; Clark & Fujimoto, 1990; Song & Montoya-Weiss, 1998). Nevertheless, influence is germane to NPD because NPD essentially addresses risk, ambiguity, and uncertainty; and is replete with functional conflicts caused by differing perceptions and self-interests (Frost & Egri, 1991; Ruekert & Walker, 1987). Similar related research results are found that the focus on the importance of understanding company NPD as a result of the efficient use of unique company capabilities that create sustained performance differentials within industries. (Barney, 1991; Rumelt, 1984; Teece, 1982; Wernerfelt, 1995). However, managers who are responsible for NPD may have limited authority to acquire information, resources, and support (Maute & Locander, 1994). Previous large-sample studies in the acquisition strategy literature have examined the motives for acquisitions and have investigated the pre-acquisition and post-acquisition performance of targeted and acquiring firms (Lubatkin, 1987; Seth, 1990; Singh & Montgomery, 1987). These studies suggest that M&A activity is likely to impact the outcomes of the NPD process. Nevertheless, despite its importance, little evidence suggests that M&A integration and NPD are related.

In co-competitive environments; among customers, competitors, suppliers and complementors, you have to compete and cooperate at the same time; the most successful strategy is to change the rules of

* Corresponding author. Tel.: +886 912 889 866.

E-mail addresses: cjs.spgb@msa.hinet.net (C.-H. Chen), ebc43874@ms32.hinet.net (Y.-Y. Chang), jameslin@cc.ncu.edu.tw (M.J. Lin).

¹ Tel.: +886 3 3412500 #3201.

the game to improve organizational competitiveness (Brandenburger & Nalebuff, 1998). Changing the game through M&A can effectively enhance the synergies in product portfolios and dynamic capabilities, and also improve NPD performance by expanding existing market boundaries. In the dynamic markets of today, companies face increasing competition, typically resulting from global competition, the energy crisis, economic changes, and turbulent or new technologies; however, these conditions also create new markets and challenges. An M&A may comprise an entry strategy for rapid growth. Enterprises undertake M&As when core competencies and resources must be enhanced for strategic purposes (Hill & Jones, 2004). Mergers and acquisitions are one of the best methods of ensuring organizational growth, completing strategic intention, and increasing core competencies.

Badwin and Clark (2000) emphasized that artifacts are no longer made by individuals and can no longer be comprehended by them. Their studies of modulization of management in the computer industry emphasize the importance of interdepartmental integration for NPD in the technology industry. Moreover, extensive research confers proposes that cross-functional participation or communication is imperative for successful NPD (Griffin & Hauser, 1996; Moenaert & Souder, 1990; Song & Parry, 1992). In 1996, Kahn also asserted that improving interdepartmental integration improves product performance.

However, many studies have examined whether the impact of interdepartmental integration on NPD performance is direct or whether it occurs via other mediating variables. An empirical study by Leenders and Wierenga (2002) demonstrated that some mechanisms impact NPD through integration; however, their study found that the direct influence of marketing-R&D integration on NPD performance is not significant ($\beta=0.15$, $t=1.70$). Therefore, they suggested further research to determine whether other variables mediate between integration and NPD performance; finally, they argued that integration impacts NPD performance through mediating variables and does not directly influence NPD performance. Langerak, Hultink, and Robben (2004) discovered that integration influences NPD performance through new product competitive advantage. Moreover, the empirical study by Swink and Song (2007) also supports the view that new product competitive advantage is positively associated with projected return on investment. Thus, new product competitive advantage is a critical mediating variable in these studies.

Product vision significantly impacts the integration process; Tassarolo (2007) emphasized the importance of product vision in the integration process and presented product vision as a moderating variable. In practice, if the staff members of a targeted firm feel that the acquiring firm has a very clear picture of a common goal, they are likely to maintain a positive working attitude because they anticipate prosperity and success in the near future. Moreover, the psychological differences between the two parties significantly impact – directly and indirectly – NPD performance. Thus, product vision is essential for generating a new product competitive advantage during the integration process.

The above literature clearly delineates the relationship between interdepartmental integration, and product vision and NPCA, and consequently, NPD performance. However, a literature review reveals four limitations in the current literature. First, although Moenaert and Souder (1990), Song and Parry (1992), Griffin and Hauser (1996), and Kahn (1996) all elucidated how interdepartmental integration impacts NPD performance, they did not focus on M&A integration. Second, some studies posited that integration does not directly impact NPD performance; rather, its effect is exerted indirectly through mediating variables (Leenders & Wierenga 2002); however, this argument has no theoretical nor empirical support from the M&A integration perspective. Third, Li and Calantone (1998) and Langerak et al. (2004) claimed that integration has an impact on NPCA and NPD performance; however, exactly how NPCA and NPD performance are

related in the M&A integration process has seldom been addressed. Finally, Tassarolo (2007) documented the moderating effects of a common goal and product vision and the integration of NPD efforts; however, it does not focus on post-M&A integration.

As mentioned in the Siemens case and relevant literature, on the basis of the program level, this study analyzes how M&A interdepartmental integration, interaction, and collaboration, all of which influence product vision and new product competitive advantage, affect NPD performance. According to current M&A trends, this study has the following objectives: (a) explore the relationship between interdepartmental integration and product vision, (b) explore the relationship between interdepartmental integration and new product competitive advantage, (c) examine the correlation between product vision and new product competitive advantage and (d) explore how product vision of team members is related to NPD performance in terms of efficiency and effectiveness during the post-M&A interdepartmental integration process. Finally this study explores how new product competitive advantage is related to NPD performance in terms of efficiency and effectiveness during the post-M&A interdepartmental integration process.

2. Literature review and hypotheses development

2.1. Strategy of mergers and acquisitions

Merger and acquisition (M&A) is the primary corporate strategy for achieving rapid growth and for overcoming obstacles to growth. Many studies have investigated the M&A integration process and M&A performance. The main schools of research are (1) the organizational behavior school, which emphasizes organizational behavior in the merging process (Mark & Mirvis, 2001), and (2) the process perspective, which discusses contingent management problems in M&A integration, which advocates that proper control measures can improve integration effectiveness after an M&A and that a good fit between synergy of a strategy and an organization can be achieved only by effectively managing M&A integration (Birkinshaw, Bresman, & Hakanson, 2000; Haspeslagh & Jemison, 1991).

An M&A can increase time efficiency, shorten time required for new product development, generate new product competitive advantage, and expand the product mix to increase NPD performance (Haspeslagh & Jemison, 1991). In recent years, many sectors of the global high technology industry, including the semiconductor and photo electronics sectors as well as the information and OEM/ODM sectors have expedited the rate of global M&A integration. Applying organizational changes through M&A and increasing product mix are one of the best methods of enhancing business growth. Brandenburger and Nalebuff (1998) argued that, in a competition and cooperation strategy, the principal concept of competition changes competition rules while changing competition rules through M&A enhances business growth. According to the business models and industry characteristics, the acquirer and acquired company integrate the core resources and capabilities. The new organization may attempt to integrate interdepartmental resources and continuously increase NPD performance after completing the re-organization which may ultimately determine the success or failure of an M&A.

2.2. Post-M&A interdepartmental integration

Integration refers to a state of shared vision, mutual goal commitments, and collaborative behaviors (Soudier, Chakrabarti, Bonoma, Avery, & Cicchinelli, 1997). Previous research in interdepartmental integration includes the following: marketing-R&D integration (Parry & Song, 1993; Ruekert & Walker, 1987; Song, Neeley, & Zhao, 1996), R&D-manufacturing integration (Adler, 1995; Ettlie, 1997; Swink, 1999; Terwiesch, Loch & DeMeyer, 2002) and marketing-manufacturing integration (Olson, Walker, Ruekert, & Bonner, 2001;

Swink & Song, 2007). However, the contributions of departments in a firm differ because of their different functions. Interactive and collaborative interdepartmental integration are typically considered the most effective means of evaluating NPD performance.

Classifying interdepartmental integration as an interactive process emphasizes the importance of departmental interaction via meetings, committees, teleconferencing, phone conversations, e-mail, and exchanging various documents (Kahn, 1996). Ruekert and Walker (1987) asserted that conflict can be eliminated and resource-utilization can be enhanced when interdepartmental dependence and interaction increase.

Meanwhile, many studies classify interdepartmental integration as interdepartmental collaboration. Examples are the extent of common consensus and the frequency of informal communication or the sharing of information and resources. The degree of cooperation between interdepartmental cooperation during both the early and later stages of a project and the subsequent activities can be measured by the following factors: frequency of communication between two departments; amount of information, advice, and technical assistance exchanged; and level of work (e.g., raw materials, specifications, and other resources) transferred (Olson et al., 2001). The concept of continuous improvement, which must be applied in inter-organizational settings, engenders the concept of collaborative improvement (Middel, Fisscher, & Groen, 2007; Millson & Wilemon, 2002). That is, the organizational environment should be collaborative rather than competitive. Companies are increasingly challenged to improve their performance and respond rapidly and effectively to market changes; involvement of all departments in post-M&A integration is essential.

The above studies and discussions of M&A and interdepartmental integration indicate that the measurement of interdepartmental integration developed by Kahn (1996) should include interactive-based and collaboration-based integration. In this study, interactive-based integration was measured using a five-item scale adapted from Schrage (1990) and Kahn (1996). Respondents were asked to evaluate the extent to which their department interacted with other departments in terms of meetings, committees, teleconferencing, phone conversations, phone mail, e-mail, and the exchange of various standard documentation. Collaboration-based integration was measured using a six-item scale developed by Kahn (1996). The respondents were asked to evaluate the degree to which their department and other departments achieved collective goals and mutual understanding, informally worked together, shared a common corporate vision, and shared ideas, information, and/or resources.

2.3. Interdepartmental integration and product vision

For a newly restructured enterprise, the extent of dependence and interaction is difficult to realize. How can these negative phenomena be improved rapidly? How can management lead a team? Most importantly, how does one build a shared vision and collective trust? Tassarolo (2007) asserted that achieving the coordination needed to process information effectively and efficiently while aligning functional perspectives with developmental goals requires a shared vision regarding new products. Product vision creates a psychologically safe environment for teams as it clearly articulates developmental goals to team members. In such circumstances, interdepartmental integration is vital because market performance following product development largely depends on the support among group members.

Some scholars have also emphasized the importance of a clear product vision to achieving superior NPD performance. Bajaj, Kekre, and Srinivasan (2004) documented the contingent effect of project oversight level, which ensures that the team pursues the right objectives, on the relationship between customer involvement and time performance. Tassarolo (2007) explored how organizational mechanisms are affected by a clearly defined and clearly communi-

cated development strategy. Kessler and Chakrabarti (1999) showed how ambiguous project directions can cause a team to pursue time-consuming and useless activities. Product vision is the inter-coordination, inter-support, and clear, stable target within a group (Lynn & Akgün, 2001). As discussed above, product vision was measured using a three-item scale adopted from Tassarolo (2007); measures were accessed where the projects had clear and formal definitions of the development objectives, which included: revenue, profits, market share, customer satisfaction, whether these objectives were clearly communicated to all the functions involved in the development, and whether there was agreement and sharing of objectives among all involved in the development. Product vision ensures a fog-free environment in terms of clarity of goals and directions in which the organizational mechanisms can help speed up new product development.

The effects of interdepartmental integration on product vision have been divided into two dimensions: interactive integration and collaborative integration. Interactive integration stresses the need for interdepartmental conferences and information exchange (Griffin and Hauser, 1996; Ruekert and Walker, 1987). Collaboration via interdepartmental integration refers to whether different departments understand each other; that is, whether they share the resources needed to achieve product competitive advantage. The above studies apparently suggest that product vision ensures clear goals and directions so that organizational mechanisms can expedite new product development. According to the abovementioned studies, product vision apparently stems from the interactive and collaborative efforts among all team member and departments. Moreover, support, coordination, inspirational message fit among organizational strategy, direction, goals, and objectives all play important roles. The success or failure of interdepartmental integration is highly dependent on these activities.

Although various empirical studies demonstrate the importance of the interactive and collaborative efforts involved in interdepartmental integration, they mainly focus on well-established enterprises. Conversely, studies of internal interdepartmental integration in M&A enterprises contribute more valuable information; moreover, these studies suggest that, in practical business administration, effective integration can help to clarify goals and the product vision. Therefore, on the basis of M&A interdepartmental integration, the following hypotheses are developed:

H1a. Interaction of post-M&A interdepartmental integration is positively correlated with product vision.

H1b. Collaboration of post-M&A interdepartmental integration is positively correlated with product vision.

2.4. Interdepartmental integration and NPCA

New product competitive advantage, as described by Langerak et al. (2004), refers to the process of providing high-quality products that are unique and superior to those of competitors; Li and Calantone (1998) also referred to the dominance of new products over similar products of the main competitor based on newness, productivity, reliability, compatibility, uniqueness, ease of use, and functionality. New product competitive advantage can be seen as a holistic conceptualization of product design quality, which encompasses the superiority or uniqueness of its features and its fitness for use; it also represents the social position of the user. New product competitive advantage can be evaluated by the following four measurements: in comparison with competitive products, whether the product offers unique features to the customer; whether the product is clearly superior to competing products in terms of meeting customer needs; whether the product quality is superior to that of competing products; and whether the product exhibits technical performance superior to

that of competing products (Song and Parry, 1997; Swink and Song, 2007).

Many studies have investigated how integration affects enterprise management through marketing, R&D, manufacturing, and perspectives of interaction and collaboration (Kahn, 1996; Kahn & Mentzer, 1998). Howell and Soucy (1990) noted that competitive advantage requires firms to strictly control manufacturing costs to ensure that their products are competitively priced. Various empirical studies have confirmed the importance of interactive integration of different departments. Particularly, Kahn (1996) demonstrated that departmental interaction positively influences NPD and product management performance. Interdepartmental collaboration can help firms win contracts, increase satisfaction, improve productivity, and improve morale and departmental confidence, all of which are essential for achieving product competitive advantage (Schrage, 1990). These research results provide evidence that product competitive advantage stems from interdepartmental integration. Hence, the following hypotheses are proposed:

H2a. Interaction of post-M&A interdepartmental integration is positively correlated with new product competitive advantage.

H2b. Collaboration of post-M&A interdepartmental integration is positively correlated with new product competitive advantage.

2.5. Product vision and new product competitive advantage

The positive effects of product vision include the ability to shape goals and directions during post-M&A integration. Enthusiasm for M&A success is typically motivated by a clear and attainable vision. In the current business environment, the success of new products depend on cooperation from suppliers, R&D, production, sales and marketing and downstream sales channels, in addition to support from upper management. Product vision is essential for integrating these resources and for providing a clear direction.

To achieve the coordination needed to process information both effectively and efficiently as well as to align functional perspectives with developmental goals, all those involved must share a strong vision (Tessarolo, 2007). Product vision elucidates directions, goals, and objectives for product development by a team (Crawford & Di Benedetto, 2003). Empirical research in product vision confirms that product competitive advantage is associated with interdepartmental integration (Crawford & Di Benedetto, 2003; Griffin & Hauser, 1996; Kahn & McDonough, 1997; Pinto, Pinto, & Prescott, 1993; Song & Parry, 1997). Product vision is generally a clear statement of goals and the direction of the company enables integration mechanisms to rapidly develop new products and ensure that progress meets the development schedule. Without product vision, an NPD team responsible for target marketing is likely to lose its direction. Interdepartmental integration efforts may eventually be offset by chaos, inefficiency, and ineffective management. This study asserts that product vision is crucial for exploiting competitive advantages to enhance NPD performance. Such sharing of a product vision creates new product competitive advantage. The enhancing effect of product vision on new product competitive advantage is greatest when team members have a clear and shared product vision. Therefore, the following hypothesis is developed:

H3. Product vision is positively correlated with new product competitive advantage.

2.6. NPD performance

2.6.1. Effects of product vision on NPD performance

Innovation effectiveness is positively, reciprocally, and strongly related to shared vision (Pearce & Ensley, 2004). Cox, Pearce, and Perry (2003) defined shared vision as a common mental model

associated with the future state of a team or as tasks that provide the basis for action within a team. Kessler and Chakrabarti (1999) proposed that ambiguous project concepts increase the speculation and conflict about what is to be produced. They claimed that this can cause time-consuming readjustments and debate; their study also empirically demonstrated that clear goals are a direct antecedent of a rapid development pace. In practice, a product vision must also be shared and supported by those involved in all NPD activities, including project team members and high-ranking officers; otherwise, participants may continually question the objectives of the team and attempt to alter the vision.

Empirical research demonstrates that assessments of NPD performance may vary according to the industry, business strategy, and design strategy of the manufacturer. As noted above, NPD performance can be categorized into two dimensions. Effectiveness of NPD performance is presented to show gains in market share, margin, customer satisfaction, and promotion of the corporate image; short-term outcomes are emphasized. Efficiency is related to market availability scheduling, successful completion of project, whether the project is on budget, and senior manager satisfaction with the project results, extending the business operation, and enhancing performance in similar competitive goods. Efficiency emphasizes long-term outcome (Lin, Chen, & Huang, 2007; Olson, Walker, Ruekert, and Robert, 1995).

Product vision typically offers a clear target and direction for enhancing integration performance. A team that is responsible for integration must have a clear product vision for efficient product development so that reasonable product development targets can be set. These improvements enable teams to increase new product development performance. To improve NPD performance, a firm must improve its integration of process, disseminate a clear product vision, and instill that vision in all those involved in the integrated NPD. Otherwise, integration mechanisms may be useless when attempting to enhance NPD performance. On the basis of the above discussion, the following hypotheses are developed:

H4a. Product vision is positively correlated with the efficiency of NPD performance.

H4b. Product vision is positively correlated with the effectiveness of NPD performance.

2.6.2. Effects of new product competitive advantage on NPD performance

Numerous studies indicate that organizational culture (Barney, 1986), total quality management (Powell, 1995), specialized supplier networks (Dyer, 1996), information technology (Powell & Dent-Micallef, 1997), and product quality (Kroll, Wright, & Heiens, 1999) are among the various factors affecting new product competitive advantage. Although these studies utilize different indices or contexts, they all affirm the positive influence of new product competitive advantage on NPD performance. Many studies report strong correlations between product success and market performance (Langerak et al., 2004; Li & Calantone, 1998; Song & Parry, 1997). Generally, products with large competitive advantage have highly innovative features and superior quality. New product competitive advantage is a facet of NPD performance that captures product desirability, quality, and associated marketing and manufacturing issues, all in terms of performance, conformance, and reliability. Given the above research findings, we propose that the influence of new product competitive advantage on NPD performance is not reversed by different situations. Therefore, the following hypotheses are proposed:

H5a. New product competitive advantage is positively correlated with the efficiency of NPD performance.

H5b. New product competitive advantage is positively correlated with the effectiveness of NPD performance.

Fig. 1 summarizes all relevant constructs in this study, their hypothesized relationships, and the theoretical framework for this study.

3. Methodology and measurement

To test these nine hypotheses, this study collected data from firms in Taiwan, China, and Hong Kong (Table 1). These countries were selected primarily because they are currently the global leading manufacturers of consumer electronics and communications products. Survey respondents were R&D and manufacturing personnel, managers, and project leaders (Table 2) employed in firms manufacturing electronic and communication products and PCs (Table 3). These industries were primarily chosen because their products have short lifecycles; this product characteristic shortens the time interval of questionnaire survey. Respondents typically held knowledge-intensive or management positions. All respondents had participated in M&A strategic planning or evaluations of post-M&A performance for at least two years; some were M&A team leaders.

3.1. Survey design and response rate

As avoiding the common method variance (CMV) bias affects validity, several measures were conducted. The first data collection phase began with five in-depth case studies of subjects with recent experience in M&A interdepartmental integration. Questionnaires were pre-tested to examine the constructs of reliability and validity and were issued by e-mail, post mail, or delivery. To examine possible non-response bias and the 'representative-ness' of the participating firms, a MANOVA analysis was performed to compare early respondents with late respondents on all of the variables. The results were not significant at $p < 0.05$; suggesting that early respondents did not significantly differ from the late respondents. Two tests were conducted to determine bias effects due to CMV: reversed items and the time-separation approach for collected data (Peng, Kao, & Lin, 2006); the time-separation approach applied to data collected in two stages with a time interval of 4–6 weeks.

3.1.1. The first stage of the questionnaire survey

The first stage of the questionnaire survey elicited basic information about the enterprise (ten items) and NPD performance (nine items). Since the study was intended to evaluate NPD performance, this survey was separated from those of other variables. Out of 1450

Table 1
Respondent characteristics (by country/territory).

Industry	Total respondent (country/territory)	Percent
China	125	49.8
Taiwan	98	39.0
Hong Kong	16	6.4
Other	12	4.8
Total	251	100

questionnaires sent, 352 were recovered. Among those recovered, thirty-one questionnaires were invalid, and 321 were effective.

3.1.2. The second stage of the questionnaire survey

In the second stage, the questionnaire gathered data regarding interdepartmental interaction and interdepartmental collaboration integration (eleven items), product vision (three items), and new product competitive advantage (six items). The 321 questionnaires were then sent to those who returned effective questionnaires during the first stage. Of the 278 questionnaires recovered, thirteen were invalid and 265 were effective. The response rate was 17.31%, and respondents included 251 M&A and fourteen non-M&A companies. This study used program level as a basis for evaluating NPD performance, and analyzed the 251 returned questionnaires on M&A.

3.1.3. Analytical tools and procedure

Subjects responded to items on a 5-point Likert scale. Both SPSS 13.0 and LISREL 8.7 were employed in analyses. To test for CMV, this study evaluated a correlated uniqueness model (Podsakoff, MacKenzi, Lee & Podsakoff, 2003). The model accounts for method effects by allowing measurement of error terms of constructs using the same method correlated in the measurement model. Comparing the correlated uniqueness model with the original measurement model identified no significant change in model fit or in any of the loading parameters, indicating that common method biases did not significantly affect data analysis results.

3.2. Statistical analysis

The research theoretical model (Fig. 1) was analyzed primarily using SEM supported by LISREL 8.7. A two-step model-building process for applying SEM was applied (Anderson & Gerbing, 1988; Hair, Anderson, Tatham & Black, 1998; Joreskog & Sorbom, 1993; Maruyama, 1998). The first step analyzes the measurement model (or

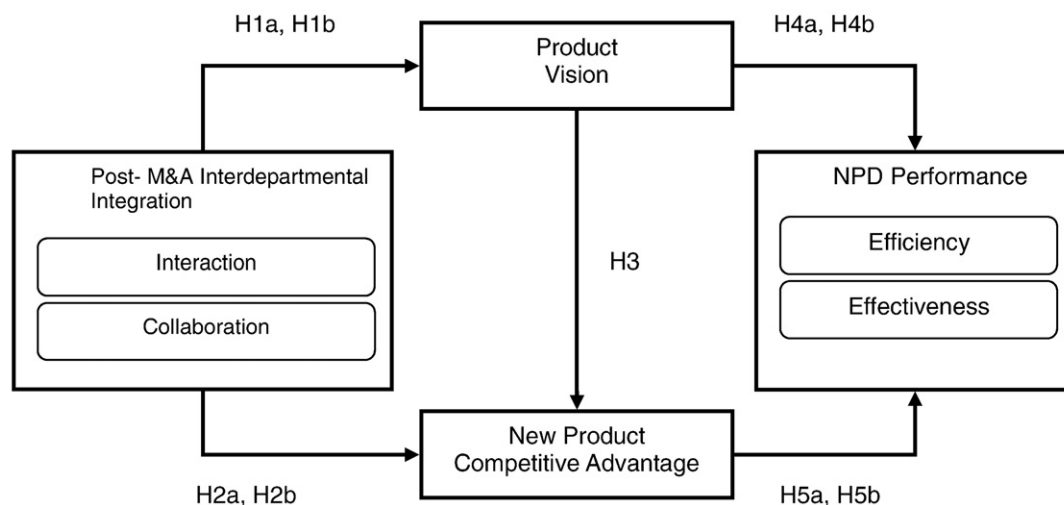


Fig. 1. Research theoretical model.

Table 2
Respondent characteristics (by industries).

Industry	Total respondent (industry)	Percent
Electronic	139	55.4
Communication	34	13.5
PC/notebook	32	12.7
Others	46	18.3
Total	251	100

the Confirmatory Factor Analysis (CFA) model), and the second step tests the structural relationships among latent constructs.

Moreover, having established an adequate measurement model, the latter step in the two-step procedure tests the hypotheses through fitted structural models for relationships among latent variables (Bollen, 1989; Kline, 1988). These tests involved comparisons with the unconstrained CFA model and tests to determine whether models with a hypothesized casual ordering among latent variables reproduced an unconstrained covariance matrix for the latent variables. Alternative models were also analyzed for model fit and interpretability.

3.3. Data analysis and results

The two-step procedure, which assesses reliability and validity of measures before their use in the full model (Anderson & Gerbing, 1988), examined the measurement model and, then, the structural model.

3.3.1. Analysis of the measurement model

This study applied CFA to test the measurement model and assess construct validity using LISREL 8.7. The six latent variables were allowed to co-vary freely in the CFA model. Parameters for all models were estimated using Maximum Likelihood (ML) with the item covariance matrix as the input. The CFA indicated that the measurement model fit the data ($\chi^2 = 331.69$; $d.f. = 309$, $\chi^2/d.f. = 1.073$, $GFI = 0.91$, $NFI = 0.96$, $CFI = 1.0$, $SRMR = 0.043$) (Fig. 2 and Table 4). The RMSEA index was 0.017 with a 90% confidence interval of 0.0–0.03. All model-fit indices exceeded their respective acceptance levels indicated by previous studies, indicating that the measurement model had acceptable fit to the collected data.

Convergent validity of the scales was verified using the three criteria suggested by Fornell and Bookstein (1982): (1) all indicator loadings should be significant and exceed 0.7; (2) construct reliability (ICR) should exceed 0.7 (Hair, Anderson, Tatham & Black, 1998); and (3) average variance extracted (AVE) by each construct should exceed the variance due to measurement error for that construct (i.e. AVE should be >0.50). In the CFA model in this study, factor loadings for almost all items exceeded the recommended level of 0.7, and all factor loadings were statistically significant at $p < 0.001$ (Table 5). Composite reliability (CR) was used to assess the internal consistency of the measurement model. Composite reliability is similar to that of Cronbach. The composite reliability of constructs in the range 0.76–

0.90 exceeded the benchmark of 0.7. The AVE was 0.52–0.63. Hence, all three conditions for convergent validity were satisfied.

Further, discriminated validity of scales was assessed using the benchmark suggested by Fornell and Bookstein (1982): the square root of the AVE from the construct should exceed the correlation between the construct and other model constructs. Table 6 lists the correlations among constructs with the AVE square root on the diagonal. All diagonal values exceeded the correlations between any construct pair. In summary, the measurement model had adequate and sufficient reliability, convergent validity, and discriminant validity.

3.3.2. Test of the structural model

All model-fit indices of the structural model exceeded their respective common acceptance levels: $\chi^2 = 334.84$; $d.f. = 313$, $\chi^2/d.f. = 1.069$, $GFI = 0.91$, $NFI = 0.96$, $CFI = 1.0$, $SRMR = 0.045$, and the RMSEA index was 0.017, suggesting that the structural model had a reasonable model-to-data fit (see Table 4).

The causal structure of the hypothesized research model reflects the assumed linear causal relationships among constructs, which were tested using a structural model. LISREL analysis was applied to the theoretical model with two exogenous latent constructs (interactive and collaboration interdepartmental integration) and four latent endogenous constructs (product vision, new product competitive advantage, NPD performance efficiency, and NPD performance effectiveness). Fig. 2 presents the resulting structural model, path coefficients and their significance, and hypothesis test results. The SEM analysis results were as follows.

Fig. 2 shows the results of SEM structural analysis. The interaction of interdepartmental integration was positively correlated with product vision ($\beta = 0.34$, $p < 0.001$), and collaboration of interdepartmental integration was also positively correlated with product vision ($\beta = 0.45$, $p < 0.001$) (Fig. 2). These analytical results supported hypotheses H1a and H2a, which state that interdepartmental integration is positively correlated with product vision. Hence, these results are consistent with theoretical expectations.

Interdepartmental integration was not significantly correlated with new product competitive advantage ($\beta = 0.17$); thus, H2a was unsupported. However, collaboration of interdepartmental integration was positively correlated with new product competitive advantage ($\beta = 0.28$, $p < 0.001$); thus, H2b was supported. Product vision was positively correlated with NPCA ($\beta = 0.22$, $p < 0.05$); Therefore, hypothesis H3 was supported. Analytical results indicated that product vision is positively correlated with the efficiency of NPD performance ($\beta = 0.35$, $p < 0.001$); thus, hypothesis H4a was supported; finally, product vision ($\beta = 0.23$, $p < 0.01$) was also positively correlated with the effectiveness of NPD performance; thus, hypothesis H4b was also supported. The analytical results supporting hypotheses H4a and H4b were consistent with theoretical expectations; that is, product vision was positively correlated with NPD performance.

Further, NPCA was positively correlated with the efficiency of NPD performance ($\beta = 0.21$, $p < 0.01$); thus, hypothesis H5a was supported; NPCA was also positively correlated with the effectiveness of NPD performance ($\beta = 0.39$, $p < 0.001$); thus, hypothesis H5b was supported. The analytical results supported hypotheses H5a and H5b, which were consistent with theoretical expectations; that is, NPCA was positively correlated with NPD performance.

The analysis confirmed that our research results confirmed most hypotheses with the exception of H2a: interaction of interdepartmental integration revealed no significant positive association with NPCA in this respect.

3.3.3. Test of the mediating effect

Regarding the mediating effect, Sobel *t*-test (Sobel, 1982) with corrected for un-normal formulation (Sampson & Breunig, 1971; also see Mackinnon, 2008, p.73) for the indirect effect among Interaction

Table 3
Respondent characteristics (by departments).

Department	Total respondent (department)	Percent
R&D	92	36.7
Manufacturing	42	16.7
Management	53	21.1
Project leader	42	16.7
Other	22	8.8
Total	251	100.0

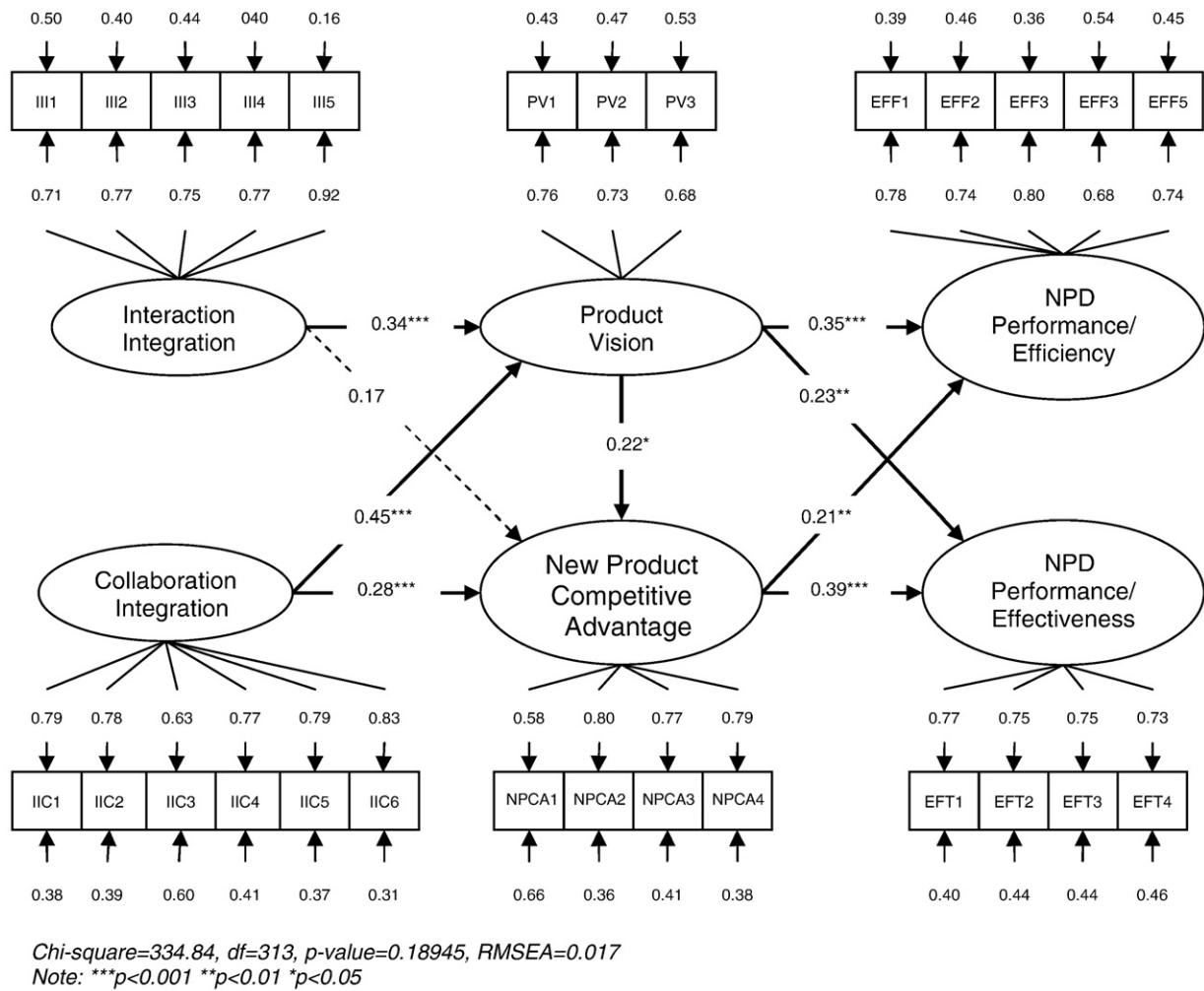


Fig. 2. SEM structural analysis.

interdepartmental integration (CFIII) → Product vision (PV) → New product competitive advantage (NPCA) was 2.02, and among Collaboration interdepartmental integration (CFIIC) → Product vision (PV) → New product competitive advantage (NPCA) was 2.12, respectively.

Beside, the *Sobel t*-test of indirect effect beginning with Interaction interdepartmental integration (CFIII) extent to the two NPD performance measures were 3.97 and 4.86 respectively (reported by LISREL). Further, the *Sobel t*-test of indirect effect beginning with collaboration interdepartmental integration (CFIIC) extent to the two NPD performance measures were 3.73, and 4.83, respectively (reported by LISREL).

All *t*-values met the criterion for statistical significance (1.96). As Table 7 illustrates, examination of mediation effect in terms of *Sobel t*-test revealed that product vision was a significant mediator of

the influence of interdepartmental integration on new product competitive advantage; new product competitive advantage was also a significant mediator of the influences of interdepartmental integration on NPD performance.

4. Discussion and managerial implications

This study has examined the possible advantages and effects of post-M&A interdepartmental integration on NPD performance using two determinants – product vision and product competitive advantage. The empirical results shed new light on how post-M&A interdepartmental integration is related to NPD performance under different contexts in the M&A integration process, which is affected by product vision and new product competitive advantage. The results are discussed below.

Table 4

Model-fit results for measurement and structural models.

Criterion statistics	($\chi^2/d.f.$)	GFI	NFI	CFI	SRMR	RMSEA
Reference	Hayduk (1987)	Bentler and Bonett (1980)	Bentler and Bonett (1980)	Bentler (1988)	Browne and Cudeck (1993)	Browne and Cudeck (1993)
Recommended value	≤ 3.00	≥ 0.90	≥ 0.90	≥ 0.95	≤ 0.08	≤ 0.05
Empirical results for the measurement model	1.073	0.91	0.96	1.0	0.043	0.017
Empirical results for the structural model	1.069	0.91	0.96	1.0	0.045	0.017

Table 5
Analysis of measurement model.

Construct and indicators	Factor loadings	t-value	CR	AVE
Interdepartmental interaction (III)				0.89 0.62
III1	0.71	12.48		
III2	0.77	14.08		
III3	0.75	13.54		
III4	0.77	14.15		
III5	0.92	18.41		
Interdepartmental collaboration (IIC)				0.90 0.59
IIC1	0.78	14.39		
IIC2	0.78	14.32		
IIC3	0.63	10.67		
IIC4	0.77	13.96		
IIC5	0.79	14.58		
IIC6	0.83	15.60		
Product vision (PV)				0.77 0.52
PV1	0.76	12.67		
PV2	0.73	12.09		
PV3	0.68	11.15		
New product competitive advantage (NPCA)				0.76 0.55
NPCA1	0.58	9.37		
NPCA2	0.80	14.13		
NPCA3	0.77	13.48		
NPCA4	0.79	13.99		
NPD performance – efficiency (NPDEFF)				0.82 0.58
NPDEFF1	0.78	13.98		
NPDEFF2	0.74	13.02		
NPDEFF3	0.80	14.45		
NPDEFF4	0.68	11.60		
NPDEFF5	0.74	13.07		
NPD performance – effectiveness (NPDEFT)				0.80 0.56
NPDEFT1	0.77	13.49		
NPDEFT2	0.75	13.07		
NPDEFT3	0.74	12.86		
NPDEFT4	0.74	12.63		

4.1. Effects of interdepartmental integration on NPCA

Regarding M&A enterprises, organizational structures and business processes are typically reformed and reengineered during M&A integration. The analytical results reveal that interaction interdepartmental integration is not positively correlated with new product competitive advantage. The following explanation is proposed: Firstly, during the early stage of post-M&A interaction interdepartmental integration, the staff of the acquired and acquiring firms are unfamiliar with each other in areas such as internal operating process and corporate culture. Consequently, meetings and other communication techniques across departments may encounter significant obstacles. Conversely, collaboration interdepartmental integration is advanced at this stage; The state is similar to that resulting from an administrative decree since there is little freedom of personnel intention; policies are followed and personnel are dedicated to creating benefits for the combined firm. Secondly, the main sample in this study was

Table 6
Correlations and AVE.

Construct	AVE	III	IIC	PV	NPCA	NPDEFF	NPDEFT
III	0.62	0.79					
IIC	0.59	0.57	0.77				
PV	0.52	0.60	0.65	0.72			
NPCA	0.55	0.46	0.51	0.50	0.74		
NPDEFF	0.58	0.28	0.32	0.46	0.38	0.76	
NPDEFT	0.56	0.25	0.40	0.42	0.50	0.54	0.75

Note: AVE > 0.50, (Hair, Anderson, Tatham & Black 1998).

Table 7
Summary of indirect effects.

	Interaction interdepartmental integration		Collaboration interdepartmental integration	
	beta	t-value	beta	t-value
Product vision	...	...	...	...
New product competitive advantage	0.07	2.02	0.09	2.12
NPD performance – efficiency	0.20	3.97	0.26	4.86
NPD performance – effectiveness	0.19	3.73	0.26	4.83

personnel in the electronics and communication industry. The new product competitive advantage of these two industries is mainly derived from product innovation and the speed of R&D; these activities are mainly performed by R&D department. Compared to administration, sales, and marketing personnel, the R&D department focuses more on developing specific targets such as new technology. Therefore, R&D personnel have little interaction with other departments. The R&D department can independently develop new techniques and continually carry out experiments until they achieve a concrete result; therefore, the interaction between colleagues and departments is not required to achieve the desired results.

The newly organized structure is confronted with post-M&A challenges related to interdepartmental integration from different corporate entities; thus, difficulties are expected. However, common product visions, directions and goals simplify collaboration between departments. In a newly restructured post-M&A firm, dependence and interaction frequency are difficult to quantify; in the fact, realizing progress of interaction interdepartmental integration is very difficult. The rapid elimination of negative phenomena and the professional management of the team are the most important issues during the M&A integration process. Moreover, to facilitate interdepartmental interaction, enhancing trust among team members is essential for diminishing these negative impacts.

4.2. The relationship between interaction and collaboration integration

The Sobel *t*-test results revealed that the total effect of collaboration interdepartmental integration far exceeds that of interaction interdepartmental integration (see Table 8). The analysis in this study suggests that staff members in acquired and acquiring firms are unfamiliar with their counterparts during the initial stages of the post-M&A integration process. In this scenario, interaction interdepartmental integration is difficult and requires considerable time, thus making it extremely difficult to create synergy.

Table 8
Summary of total effects.

	Interaction interdepartmental integration		Collaboration interdepartmental integration	
	beta	t-value	beta	t-value
Product vision	0.36	4.23	0.46	5.45
New product competitive advantage	0.22	2.85	0.34	4.23
NPD performance – efficiency	0.20	3.97	0.26	4.86
NPD performance – effectiveness	0.19	3.73	0.26	4.83

This phenomenon suggests that all managers involved in interdepartmental integration should attempt to facilitate positive interaction between employees through activities and events such as product exhibitions within the enterprise, formal or informal interdepartmental get-togethers, and dissemination of cultural knowledge for employees from different countries and races. Such efforts can substantially reduce the so-called mapping time. When employees unfamiliar with each other and develop a tacit understanding of each other's workings, the ability of the team to develop new products is diminished. This situation is exacerbated in transnational corporations.

Managers should be extremely careful when addressing minor issues arising from interaction between the acquiring and acquired firms. Although the problems may appear minor, they may cause dilemmas or conflicts that adversely affect harmony within the new organization. As exhaustion during post-M&A integration increases, NPD performance decreases. However, effective communication among departments yields many positive outcomes such as stimulating discussions, mutual understanding, collaboration, and sharing; these outcomes then increase the effectiveness and efficiency of NPD performance.

4.3. Mediating effects of product vision and NPCA

Technology can continuously improve quality of life given an appropriate vision and goals. The continuous pursuit of a noble vision has led to some of the greatest achievements in human history. Empirical results show that product vision is positively correlated with new product competitive advantage; that is, the results support hypothesis H3. This study corroborates the view that, when an ambitious product vision is formulated, improved new product competitive advantage enhances NPD performance, which heavily relies on adequate post-M&A interdepartmental integration. These analysis results also reveal that product vision is crucial for enhancing new product competitive advantage as it relates to increased NPD performance; thus, the results support H4a and H4b. An effectively disseminated product vision clarifies the goals and the direction required for the rapid development of new product competitive advantage and NPD performance. Hence, managers with high expectations of achieving improved NPD performance should adopt integration mechanisms and share clear objectives and directions with those involved in NPD; that is, managers should consider not only customers, but also personnel within the firm (i.e., team members, functional representatives, and high-ranking managers) and outside the firm (i.e., suppliers).

4.4. Managerial implications

Analysis results reveal that the average impact of collaboration interdepartmental integration is much larger than that of interaction interdepartmental integration (Table 8), and, similarly, interaction interdepartmental integration is not significantly correlated with new product competitive advantage although collaboration interdepartmental integration is significantly correlated with new product competitive advantage. This finding is most likely due to the difference in corporate culture between the acquiring and acquired firms as well as the operations of management may have originally pertained to different departments prior to M&A integration; in the early stages of post-M&A integration, the personnel of the two firms or different departments is unfamiliar with the working style and thinking of their counterparts, and hence, effective interaction is difficult. In order to achieve high management performance, we suggest enforcing administrative decrees during the early stages of interdepartmental integration with other accompanying managerial strategies, and then gradually reducing the administrative decrees after achieving the desired level of interaction.

In addition to support from top-level management, the success of NPD depends on the cooperation among suppliers; customers; downstream sales channels; and internal departments such as R&D, production, sales, and marketing. In an M&A, the acquiring firm usually makes a substantial investment to obtain the resources or core competencies of the acquired firm; in this condition, the success or failure of this interdepartmental integration in this period is particularly important. Product vision is essential for integrating these resources and mobilizing them toward different functions or a specific direction. Enterprise vision influences staff morale and motivation; similarly, product vision influences new product competitive advantage and NPD performance.

As abovementioned, the enterprise strategies determine the objective of the M&A and the direction of the post-M&A integration. Under a team driven by product vision, the enterprise can obtain better new product competitive advantage, and then improve NPD performance. This affirms the importance of corporate strategies, the integration process, and product vision in M&As. Product vision is particularly important to managers responsible for NPD as it can aid the managers in the effective management of M&As, and in their efforts to achieve outstanding performance in the face of global competition and a dynamic economic business cycle.

5. Limitations and opportunities for future research

Some limitations of this study suggest future research directions. First, as this research focused on M&A enterprises, we suggest that future studies compare M&A and non-M&A NPD performance. Such a comparison can provide significant contributions and recommendations for enterprises intending to merge with other enterprises. The subjects of this study were enterprises that have undertaken M&A activities. This study, which has few samples of non-M&A firms, compared those with enterprises in different industries that have experienced M&As. This study identified significant differences between M&A and non-M&A NPD performance as it relates to interdepartmental integration. Although the non-M&A sample size was too small to statistically support this finding, further investigation is warranted.

Regarding different industries, different industrial products have different life cycles, and enterprise product visions and integration strategies differ as well. Further studies investigating these topics would prove helpful to enterprises evaluating the effectiveness of M&As. Additionally, Kitching (1967) identified five general acquisition types: vertical integration, horizontal integration, concentric marketing, concentric technology and conglomerates. Different M&A types may have different integration effects, and these directions deserve further investigation.

Further research into interdepartmental integration is particularly important given the current in M&A activity and the use of product vision approaches for generating new product competitive advantage and NPD performance. If problems persist during interdepartmental activities, further effects that instruct departments to work closely would undoubtedly increase frustration and debate among the involved departments, which would eventually reduce NPD performance.

Appendix A. Measurement items

Details of the survey

In this study, a “new product” refers to “improvements in an existing product, improvements in the functions of a product, and the development of new products for the company.” Among all the NPD projects that you have participated in, please choose the option that best represents your case from the options given below.

Please ensure that the project meets the following conditions: the development process of the product is complete, and the products are officially on sale after post-M&A integration. Further, with regard to this product, answer the following questions about the actual situation of your company.

Merger and acquisition (M&A)

The following question is about M&A integration. Please choose the option that best represents your case from the options given below.

1. Has your company had acquisition/merger in recent two years? If there is no acquisition/merger, please skip items 2–5 and go to next section.
2. Are the two parties in acquisition/merger in “the same industry”?
3. Are the two parties in acquisition/merger the upstream and downstream companies?
4. Will “product mix” increase after acquisition/merger?
5. Does it aim to acquire professional or core techniques (such as patent...)?

Interdepartmental integration

The definition of interdepartmental integration adapted from Kahn (1996) encompasses interactive-based and collaboration-based integration.

1. Interactive-based integration was measured using the five-item scale adapted from Van de Ven, Walker and Liston (1979), and Kahn (1996). Respondents were asked assess the degree to which their department interacted with other departments via meetings, committees, teleconferencing, phone conversations, phone mail, e-mail, and the exchange of standard documentation.

For the team members involved in post-M&A interdepartmental integration, the purpose of the questions listed below is to estimate the level of understanding what degree did your department interact with other department in regards to the below activities. Please choose the option that best represents your case from the options given below.

(1 = never; 2 = seldom; 3 = occasionally; 4 = often; 5 = quite frequency):

- Meetings (III1)
- Committees/Task forces (III2)
- Phone conversation (III3)
- Phone mail and e-mail (III4)
- Exchanges of standard documentation (III5).

2. Collaboration-based integration was measured using a six-item scale developed by Kahn (1996). Respondents were asked to assess the degree to which their department and other departments achieved collective goals, had mutual understanding, informally worked together, shared the same company vision, and shared ideas, information, and/or resources.

For the team members involved in post-M&A interdepartmental integration, the purpose of the questions listed below is to estimate the level of understanding what degree did your department had the collaboration interdepartmental integration with the other department. Please choose the option that best represents your case from the options given below.

(1 = never; 2 = seldom; 3 = occasionally; 4 = often; 5 = quite frequency):

- Achieve goals collectively (IIC1)
- Have a mutual understanding (IIC2)
- Informally work together (IIC3)
- Share ideas, information, and/or resources (IIC4)
- Share the same vision for the company (IIC5)
- Work together as a team (IIC6).

Product vision

This construct was measured using a three-item scale adopted from Tassarolo (2007). Respondents were asked to access the degree (1 = Never; 2 = Seldom; 3 = Occasionally; 4 = Often; 5 = Quite Frequency):

For the team members involved in post-M&A interdepartmental integration, the questions listed below attempt to estimate the understanding of product vision in team members during the NPD process. Please choose the option that most closely represents your case from the options given below.

1. Where the projects had clear and formal definitions of development objectives, including, but not limited to, revenues, profits, market share, customer satisfaction (PV1).
2. Whether these objectives were clearly communicated to all involved in the product development (PV2).
3. Whether an agreement existed and objectives shared among those involved in the product development (PV3).

New product competitive advantage

New product competitive advantage was measured based on the evaluation of the following four measurements (Song and Parry, 1997; Swink and Song, 2007): compared to competitive products (1 = never; 2 = seldom; 3 = occasionally; 4 = often; 5 = quite frequency):

1. This product offered some unique features (NPCA1).
2. This product was clearly superior to competing products in terms of meeting customer needs (NPCA2).
3. This product had superior quality compared with that of competing products (NPCA3).
4. This product had superior technical performance compared to that of competing products (NPCA4).

NPD performance

The definition of NPD performance adapted from Olson et al. (1995); Lin et al. (2007), encompasses NPD efficiency and NPD effectiveness (1 = never; 2 = seldom; 3 = occasionally; 4 = often; 5 = quite frequency).

1. Efficiency of NPD performance is a measure of gains in target market share, target margin, customer satisfaction, and enhancement of corporate image.
 - Becoming available in the market on schedule (NPDEFF1)
 - Completing the project successfully in budget (NPDEFF2)
 - The senior managers satisfy the project results (NPDEFF3)
 - Extending business operation (NPDEFF4)
 - Getting better performance in a lot of similar and competitive goods (EFF5).
2. Effectiveness of NPD performance is related to market availability of the product on schedule, successful completion of a project, whether a project is on budget, senior manager satisfaction concerning the project results, extending business operations, and improving performance in similar and competitive goods.
 - Gaining the targeting market share (NPDEFT1)
 - Gaining the targeting margin (NPDEFT2)
 - Customer satisfaction (NPDEFT3)
 - Promoting the corporate image (NPDEFT4).

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Chi-Hsiang Chen received his PhD of Business Administration from the National Central University, Taiwan, in 2009. Dr. Chen has over 15 years of high technology products marketing experiences. His current research focuses on technology intensive enterprise M&A, post-merger integration and high technology product portfolio management.

Dr. Yong-Yang Chang is chairman of the Department of Business and Entrepreneurial Management, Kainan University in Taiwan. Dr. Chang has over 30 years of financial management experiences. He has extensive financial consulting experiences in leading organization globally, including M&A activities. His current research focuses on entrepreneurial management, technology intensive enterprise innovations, reforms and financial management.

Dr. Ming-ji James Lin is a Professor of the Department of Business Administration at the National Central University in Taiwan. He received his PhD in Business Administration from Chingchi University in 1992. He was a visiting scholar at Harvard University from 1995 to 1996, and Yoshida Hideo Memorial Foundation from 2002 to 2004. His areas of expertise include management of R&D, innovation, technology transfer and NPD management.